WORKED SOLUTION

STEP I 1993, Question 5

Pure • Modulus and Triangle Inequality

STEP I 1993, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q5

The key idea

Everything rests on the triangle inequality: the length of a sum is at most the sum of the lengths. Applied term by term, it bounds the polynomial by a geometric series in $|z|$, and with $|z| \leq \frac{1}{2}$ that series stays strictly below 1. A root must make the polynomial part equal to -1, which has modulus exactly 1, so a root can't lie in the disc $|z| \leq \frac{1}{2}$.

Solution

The modulus and the product rule

For $z = x + \mathrm{i}y$ with x, y real, the modulus is the distance from the origin to the point (x, y):

$$|z| = \sqrt{x^2 + y^2} \qquad \text{(the non-negative square root)}.$$

Let $z_1 = a + \mathrm{i}b$ and $z_2 = c + \mathrm{i}d$ with a, b, c, d real. Then $z_1 z_2 = (ac - bd) + \mathrm{i}(ad + bc)$, so

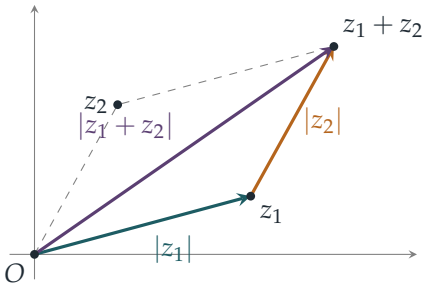
$$\begin{aligned} |z_1 z_2|^2 &= (ac - bd)^2 + (ad + bc)^2 \\ &= a^2c^2 - 2abcd + b^2d^2 + a^2d^2 + 2abcd + b^2c^2 \\ &= a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2 \\ &= (a^2 + b^2)(c^2 + d^2) = |z_1|^2 |z_2|^2. \end{aligned}$$

Both $|z_1 z_2|$ and $|z_1||z_2|$ are non-negative, so taking square roots gives $|z_1 z_2| = |z_1||z_2|$, as required.

The triangle inequality

In the Argand diagram, let O be the origin, A the point z_1 and C the point $z_1 + z_2$. Then $\overrightarrow{AC}$ represents z_2 (it is the same vector as O to z_2), so OAC is a triangle with side lengths

$$OA = |z_1|, \qquad AC = |z_2|, \qquad OC = |z_1 + z_2|.$$



A straight line is the shortest route between two points: going from O to C directly is no longer than going via A. So

$$|z_1 + z_2| \leq |z_1| + |z_2|,$$

with equality exactly when O, A, C are collinear with A between them, that is, when z_1 and z_2 point in the same direction (or one of them is 0).

If you prefer algebra: writing $\bar{z}$ for the conjugate,

$$\begin{aligned} |z_1 + z_2|^2 &= |z_1|^2 + |z_2|^2 + 2\,\mathrm{Re}(z_1\bar{z}_2) \\ &\leq |z_1|^2 + |z_2|^2 + 2|z_1||z_2| = (|z_1| + |z_2|)^2, \end{aligned}$$

since the real part of a complex number is at most its modulus, and $|z_1\bar{z}_2| = |z_1||z_2|$ by the product rule.

Applying the inequality repeatedly (an easy induction on the number of terms) gives

$$|w_1 + w_2 + \cdots + w_n| \leq |w_1| + |w_2| + \cdots + |w_n|$$

for any complex numbers $w_1, \ldots, w_n$.

The bound on the polynomial

Suppose $|a_j| \leq 1$ for each j and $|z| \leq \frac{1}{2}$. Then

$$\begin{aligned} \left|a_n z^n + a_{n-1}z^{n-1} + \cdots + a_1 z\right| &\leq |a_n||z|^n + |a_{n-1}||z|^{n-1} + \cdots + |a_1||z| \\ &\leq |z|^n + |z|^{n-1} + \cdots + |z| \\ &\leq \left(\tfrac{1}{2}\right)^n + \left(\tfrac{1}{2}\right)^{n-1} + \cdots + \tfrac{1}{2} \\ &= 1 - \left(\tfrac{1}{2}\right)^n \\ &< 1. \end{aligned}$$

The first line is the n-term triangle inequality together with the product rule (used repeatedly, so that $|a_k z^k| = |a_k||z|^k$). The second uses $|a_j| \leq 1$, the third uses $0 \leq |z| \leq \frac{1}{2}$, and the fourth sums the finite geometric series. This is the required strict inequality.

Ask yourself Where does the *strict* inequality come from? Each step above is only "$\leq$".

It comes from the finite geometric series: $\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2^n}$ falls short of 1 by exactly $\frac{1}{2^n}>0$. That gap is what the deduction needs.

The roots

(The paper's final line says $|x|>\frac{1}{2}$; this is evidently a misprint for $|w|>\frac{1}{2}$, since w is the root being discussed.)

Let w be a root, so

$$a_n w^n + a_{n-1}w^{n-1} + \cdots + a_1 w = -1, \qquad \text{and hence} \qquad |a_n w^n + \cdots + a_1 w| = |-1| = 1.$$

Suppose, for contradiction, that $|w| \leq \frac{1}{2}$. Then the previous part, with $z = w$, gives $|a_n w^n + \cdots + a_1 w| < 1$, which contradicts the modulus being exactly 1. Therefore every root satisfies $|w| > \frac{1}{2}$.

Checks

- The bound is attained in the limit: with every $a_j = 1$ and $z = \frac{1}{2}$ the left-hand side is exactly $1 - 2^{-n}$.
- The root result is sharp too. Taking every $a_j = -1$ gives $1 - z - z^2 - \cdots - z^n = 0$, whose positive root is 1 for $n = 1$, 0.618 for $n = 2$ (the golden ratio conjugate), 0.5087 for $n = 5$ and 0.50025 for $n = 10$: always above $\frac{1}{2}$, but tending to it.
- Numerically, we tested 50 000 random choices of a_j in the unit disc and z in the disc of radius $\frac{1}{2}$ (largest value found 0.79), and found all roots of 2000 random such polynomials: the smallest root modulus was 0.562.

Common slips

- Writing $|a_1 z + \cdots + a_n z^n| = |a_1||z| + \cdots + |a_n||z|^n$. The triangle inequality only gives "$\leq$".
- Using the infinite sum $\frac{1}{2}+\frac{1}{4}+\cdots = 1$, which gives only ≤ 1 and spoils the deduction. Use the finite sum.
- Forgetting to justify $|z^k| = |z|^k$: it follows from the product rule you have just proved.

Beyond STEP

This is a baby version of the bounds on polynomial roots that underpin numerical root-finding: Cauchy's bound puts every root of a polynomial in a disc whose radius

depends only on the sizes of the coefficients, and the same "dominant term wins" argument drives Rouché's theorem in complex analysis. The triangle inequality itself is the defining property of a *metric*, the abstract notion of distance at the heart of first-year analysis. For a beautifully geometric treatment of all this, read Needham, *Visual Complex Analysis*.

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